

Analysis group meeting

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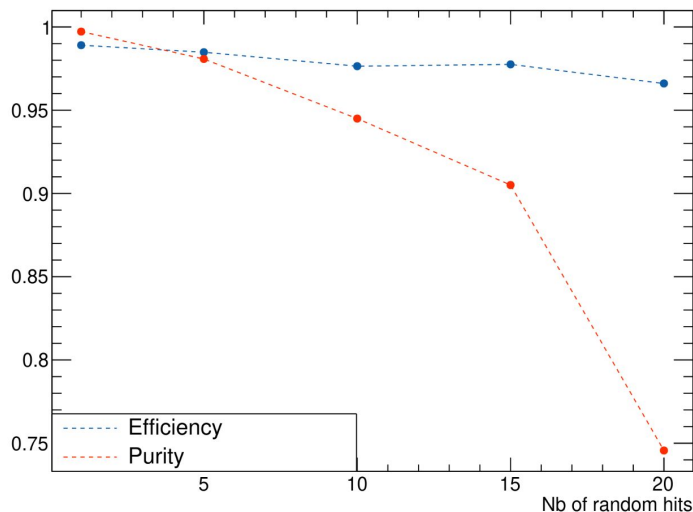
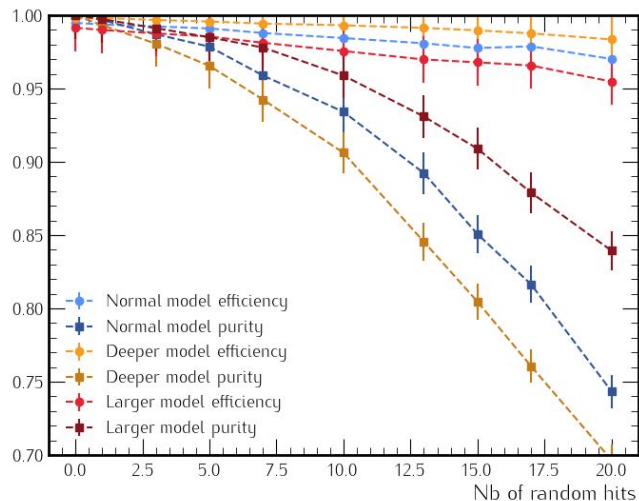
17 December, 2024



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AI for ALERT

- Implementation in COATJAVA is done.
- Multithread (with CLARA) is working well.
- Got the efficiency and purity from the COATJAVA implementation
- Made a pull request to mpaolone github repo



- Dan asked me to write a note about my analysis on DC occupancy and FT dose for ALERT.
- Raffaella and Florian need to agree with my conclusions.
- I have written it.

Data: pass0v7 target: LD2

- Electron cuts:

- $\text{pid} = 11$ and $\text{status} < 0$
- $-5 < \text{chi2pid} < 5$
- $-15 < v_z < 5 \text{ cm}$

- Pions cuts:

- $\text{pid} = 211, -211$
- $-5 < \text{chi2pid} < 5$
- $-15 < v_z < 5 \text{ cm}$

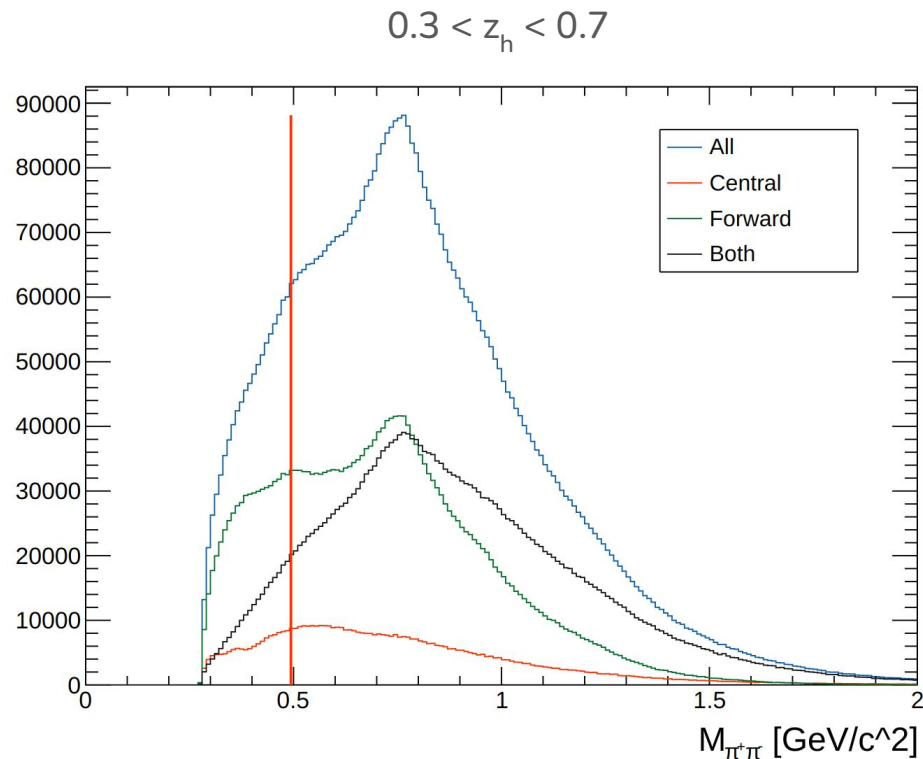
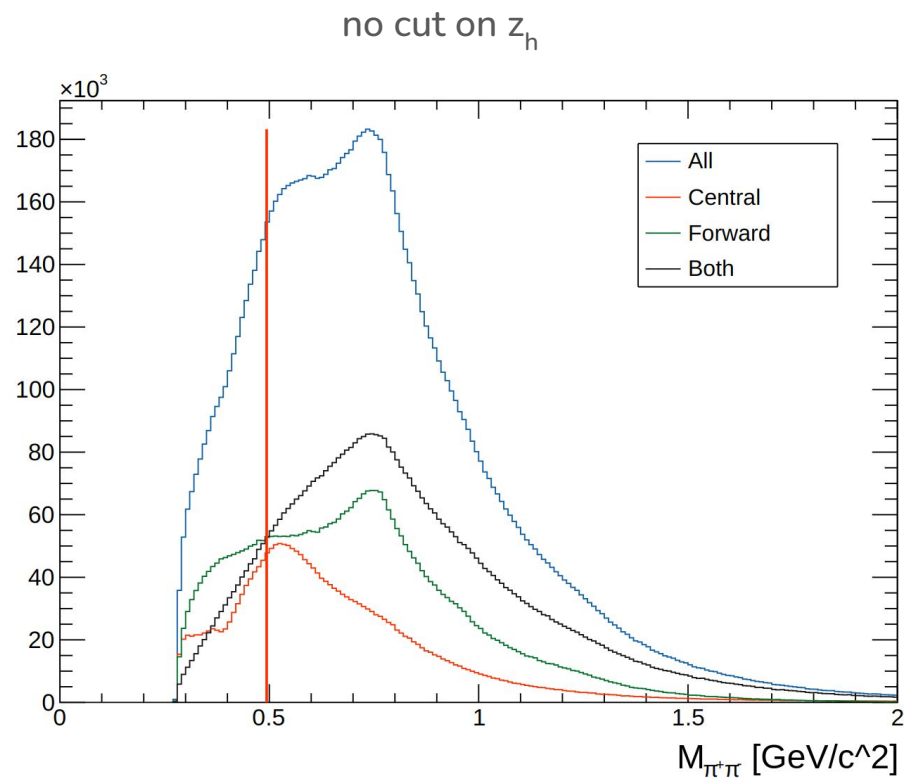
- Kinematical cuts:

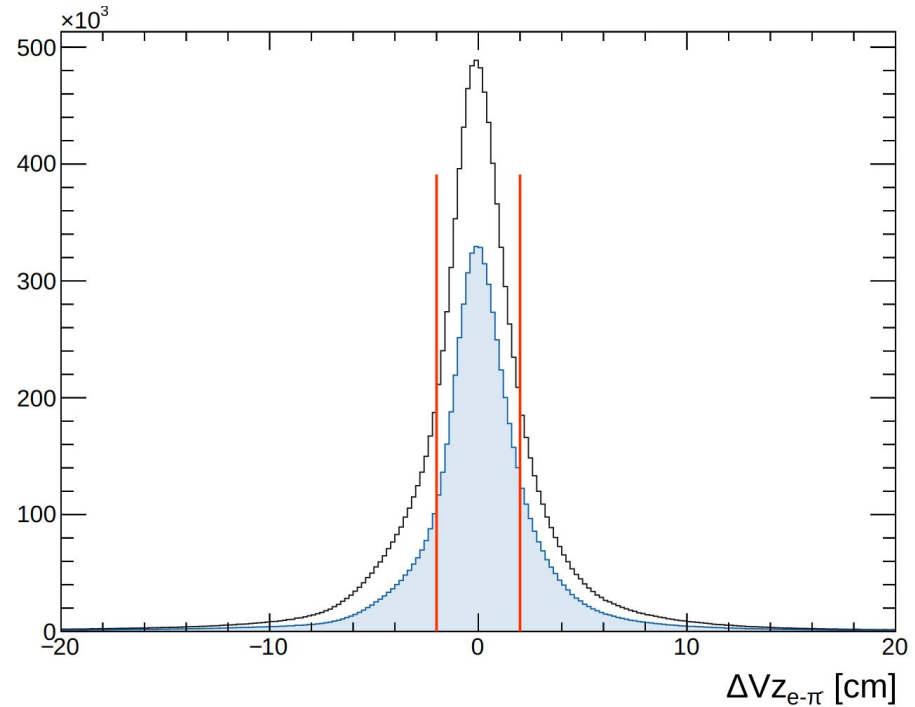
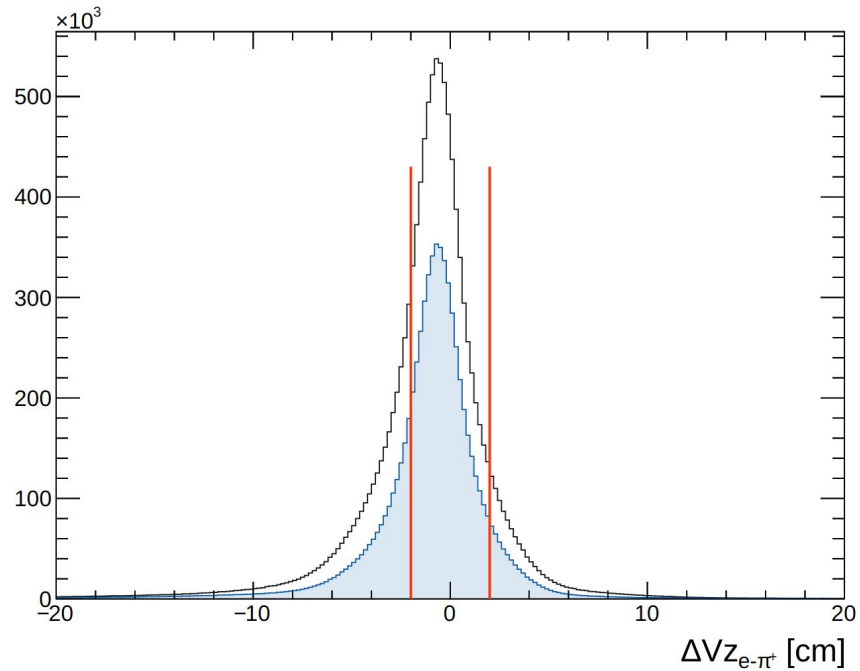
- $Q^2 > 1 \text{ GeV}^2$ to select the DIS regime
- $W > 2 \text{ GeV}$ to avoid the resonance region
- $y < 0.85$ to avoid large contribution from radiative processes

Reason for a cut on z_h :

lower limit: to reduce the probability of hadrons originating from the target fragmentation region

higher limit: to minimize the contribution of exclusive channels (could be changed to a cut on the missing mass of the ehX system(will not work for nuclear target))

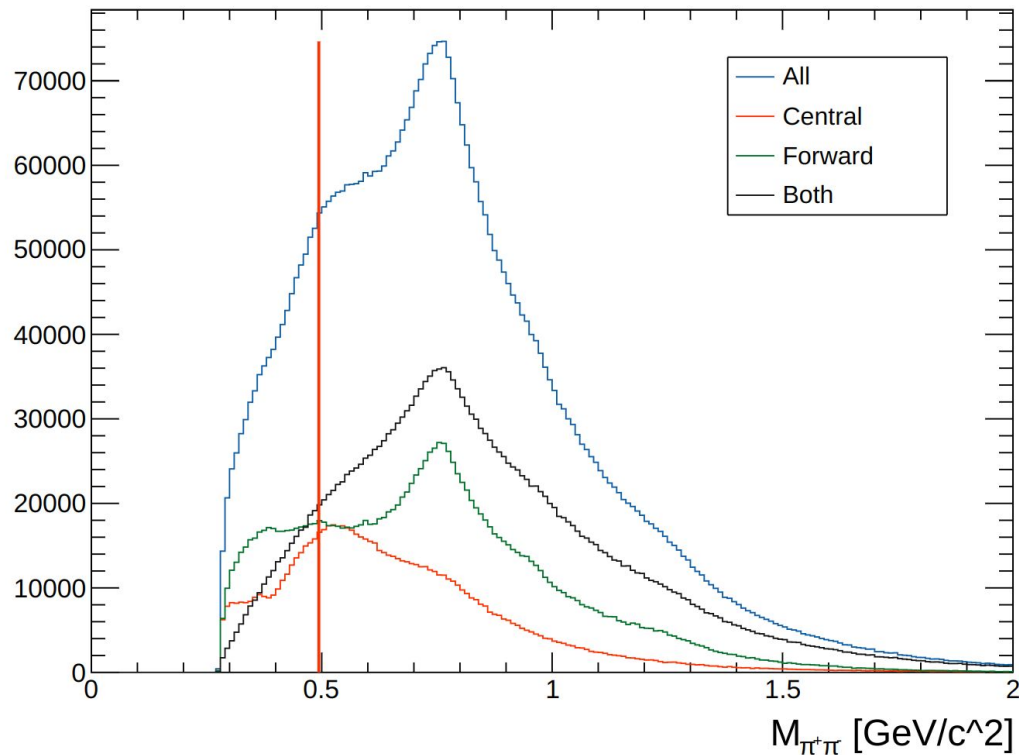




Add a cut on deltavz_e_pp and deltavz_e_pm :

$$-2 < \Delta V_{z_{e\pi^+}} < 2 \text{ cm}$$

$$-2 < \Delta V_{z_{e\pi^-}} < 2 \text{ cm}$$

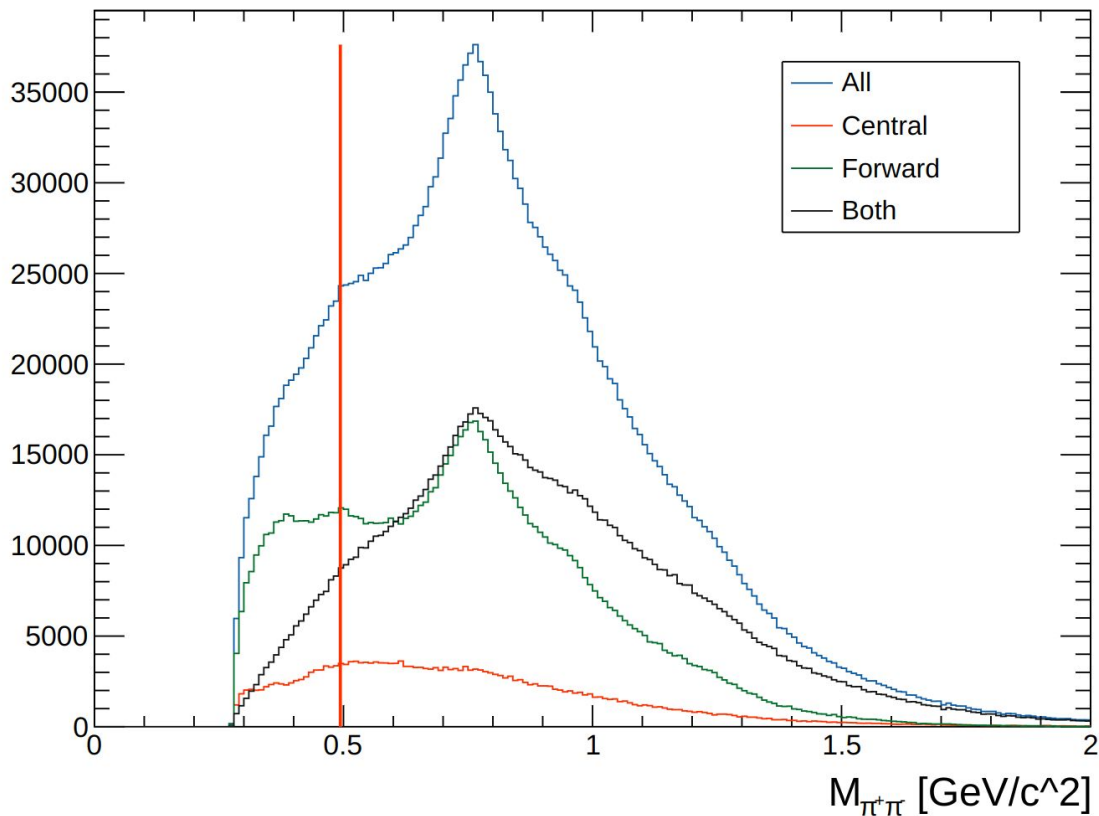


Add a cut on $\Delta V_{z_{e\pi^+}}$ and $\Delta V_{z_{e\pi^-}}$:

$$-2 < \Delta V_{z_{e\pi^+}} < 2 \text{ cm}$$

$$-2 < \Delta V_{z_{e\pi^-}} < 2 \text{ cm}$$

$$0.3 < z_h < 0.7$$



- RG-D study:

- Vz sector depend fit Pion plus/minus with Sn and Cu target cross check with Matt
- Work on the cross check for the nuclear transparency vs Q^2 and vs l_c
- Work on K^0 analysis, improve K^0 identification with detached vertex, compute multiplicity ratio and asymmetry, remove vz cut
- Daniel (and Raphael) want to produce simulation files and want to know where they should store the files. He wants to know if there is an “official” location on the farm where they can be stored.

- AI for ALERT

- Redo the efficiency/purity analysis using luminosity simulation to have a better background
- Check how often a hit from GEMC come from a secondary particle (Whitney)
- Comparison between conventional and AI track finding
- Compute the errors bar of the efficiency and purity using Poisson distribution